INTERNSHIP PROJECT REPORT

ElevateLabs CyberSecurity Internship | Project 4

Project title	Password Strength Analyzer with Custom Wordlist Generator
Intern	Ameeth Paul
Technology	Python | Kali Linux | zxcvbn | argparse | Git and GitHub

1. Abstract

This project implements a Python command-line tool that assesses password strength and generates controlled custom wordlists for authorised security testing. The analyzer uses the zxcvbn password-strength estimator together with additional checks for password length, common passwords, and meaningful username matches. The wordlist generator applies a limited set of transformations, removes duplicate candidates, enforces a maximum of 10,000 candidates per run, and exports the results to a plain-text file.

2. Introduction

Weak and predictable passwords can expose user accounts to guessing attacks. Passwords based on common words, personal information, repeated patterns, or simple substitutions may be easier to guess than users expect. The project was developed to demonstrate password-strength assessment and to show how predictable password variations can be generated in a controlled environment. The application is a CLI implementation; a graphical interface is not part of the current deliverable.

3. Tools and Technologies Used

- **Python 3.13:** Main programming language.
- **zxcvbn 4.5.0:** Estimates password guessability by recognising common patterns and structures.
- **argparse:** Python standard-library support for command-line argument handling.
- **getpass:** Supports hidden password entry in the terminal.
- **Kali Linux:** Development and testing environment.
- **Git and GitHub:** Version control, source-code hosting, and project evidence management.

Implementation note: NLTK was listed in the internship mini-guide but was not used as a dependency in the completed implementation.

4. Steps Involved

- Created the Python project structure and an isolated virtual environment, then installed zxcvbn.
- Implemented password analysis using zxcvbn and project-specific rules for minimum length, common-password matches, and username-related weaknesses.
- Made username input mandatory and rejected blank or whitespace-only values.
- Applied classification rules: zxcvbn scores 0–1 map to Weak, 2 to Moderate, and 3–4 are candidates for Strong subject to additional requirements. Strong requires at least 14 characters; a meaningful username match prevents a Strong result.
- Implemented a custom wordlist generator with supported leetspeak substitutions and suffix patterns, including !, @, #, 123, 123!, and !123.
- Added candidate deduplication and a maximum output limit of 10,000 candidates per run.
- Added output-path validation and overwrite protection before exporting generated candidates as a .txt file.
- Documented the implementation and tests, maintained an evidence register, captured seven test screenshots, and pushed the project files and evidence to GitHub.

5. Testing and Results

Testing covered mandatory username validation, username-match classification behaviour, common-password and length checks, suffix-input validation, duplicate handling, the 10,000-candidate limit, text-file export, invalid output paths, and overwrite protection. Seven screenshots were committed in the repository as supporting evidence. The project also maintains a test plan, engineering log, and evidence register.

6. Conclusion

The project delivers the two principal functions requested in the assignment: password strength analysis and custom wordlist generation with text-file export. It demonstrates Python programming, input validation, file handling, dependency isolation, testing, and version control. The tool provides an educational estimate rather than a guarantee of password security. It does not crack passwords, attempt account logins, or check external breach databases. Passwords and usernames are not intentionally persisted or logged by the application. Use of generated wordlists is limited to authorised security testing.

GitHub repository: ElevateLabs Project 4 - Password Strength Analyzer